
CS 592: Reactive Design Patterns

Course Code	CS 592
Course Name	Reactive Design Patterns
L-T-P-C	2-0-0-2
Prerequisites	Instructor's permission
Intended for	UG and PG
Distribution	Elective for BTech CSE, MS, PhD
Approval	15th BoA (Dec 2016); 12th Senate (Mar 2017)

Course Contents

- Review of Multicore processor, cache & memory hierarchy and cache coherency. [2 Lectures]
- Review of Process & thread. [1 Lecture]
- Thread Pool & their scheduling & User Mode scheduling. [3 Lectures]
- Event and message communication & their context & handling. [1 Lecture]
- Asynchronous Communication, Promise & Future. [2 Lectures]
- Method execution as an independent thread & Green Threads. [2 Lectures]
- Message Flow Patterns: Request-Response & Ask. [4 Lectures]
- Flow Control Patterns: Pull, Managed Queue, Drop. [3 Lectures]
- Replication Patterns: Active-Reactive, Active-Active. [4 Lectures]
- Resource Management Patterns: Resource Pool & Complex Command Pattern. [3 Lectures]
- Mutability & Synchronization primitives. [2 Lectures]
- Software Transactional Memory. [4 Lectures]